**UNIDAD DE INGENIERÍA EN COMPUTACIÓN**
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**Práctica 01 (%)** Fecha de Entrega: 6 de mayo 2026

# Sistemas de Numeración

Nota de la Tarea: Se debe realizar los 10 primeros ejercicios de cada sección (del 1 al 10), los demás ejercicios quedan para práctica adicional.

## 1 Conversión de Sistemas de Numeración

Realice las siguientes conversiones con desarrollo paso a paso y dejando espacio para resolución.

Convertir a base 10:

1. $11010111_2 = 215_{10}$
2. $3F7A_{16} = 16\,250_{10}$
3. $5412_8 = 2826_{10}$
4. $1010110_2 = 86_{10}$
5. $7E_{16} = 126_{10}$
6. $302_8 = 194_{10}$
7. $11011001_2 = 217_{10}$
8. $5B_{16} = 91_{10}$
9. $745_8 = 485_{10}$
10. $11100011_2 = 227_{10}$
11. $4D_{16}$
12. $611_8$
13. $100110_2$
14. $FF_{16}$
15. $33_8$
16. $100001_2$
17. $AD_{16}$
18. $110_8$

# 1: Sistemas de Numeración:

## A. Convertir a base 10:

1) $1101\ 0111_2 = 215_{10}$

| $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|---|---|
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |

$\therefore 128 + 64 + 16 + 4 + 2 + 1 = 215_{10}$

2) $3F7A_{16} = 16\ 250_{10}$

F = 15
A = 10

$3 \cdot 16^3 = 12\ 288$
$15 \cdot 16^2 = 3840$
$7 \cdot 16^1 = 112$
$10 \cdot 16^0 = 10$

$\therefore 12\ 288 + 3840 + 112 + 10 = 16\ 250$

3) $5412_8 = 2826_{10}$

$2 \cdot 8^0 = 2$
$1 \cdot 8^1 = 8$
$4 \cdot 8^2 = 256$
$5 \cdot 8^3 = 2560$

$\therefore 2560 + 256 + 8 + 2 = 2826_{10}$

4) $1010110_2 = 86_{10}$

| $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|---|---|
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 |

$\therefore 64 + 16 + 4 + 2 = 86_{10}$

5) $7E_{16} = 126_{10}$

E = 14

$14 \cdot 16^0 = 14$
$7 \cdot 16^1 = 112$

$\therefore 112 + 14 = 126_{10}$

6) $302_8 = 194_{10}$

$2 \cdot 8^0 = 2$
$0 \cdot 8^1 = 0$
$3 \cdot 8^2 = 192$

$192 + 0 + 2 = 194_{10}$

7) $1101\ 1001_2 = 217_{10}$

| $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|---|---|
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |

$\therefore 128 + 64 + 16 + 8 + 1 = 217_{10}$

8) $5B_{16} = 91_{10}$

B = 11

$11 \cdot 16^0 = 11$
$5 \cdot 16^1 = 80$

$\therefore 80 + 11 = 91_{10}$

9) $745_8 = 485_{10}$

$5 \cdot 8^0 = 5$
$4 \cdot 8^1 = 32$
$7 \cdot 8^2 = 448$

$\therefore 448 + 32 + 5 = 485_{10}$

10) $1110\ 0011_2 = 227_{10}$

| $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|---|---|
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |

$\therefore 128 + 64 + 32 + 2 + 1 = 227_{10}$

19. $11111111_2$
20. $2A_{16}$

## 2 Conversión de Decimal a Otras Bases

Convierta los siguientes números decimales a binario, octal y hexadecimal.

1. $237625_{10} = 0011\ 1010\ 0000\ 0011\ 1001_2 = 720071_8 = 3A039_{16}$
2. $153375_{10} = 0010\ 0101\ 0111\ 0001\ 1111_2 = 453\ 437_8 = 2571F_{16}$
3. $77_{10} = 0000\ 1101_2 = 115_8 = 4D_{16}$
4. $4205_{10} = 0001\ 0000\ 0110\ 1101_2 = 10\ 155_8 = 106D_{16}$
5. $255_{10} = 1111\ 1111_2 = 377_8 = FF_{16}$
6. $51225_{10} = 1100\ 1000\ 0001\ 1001_2 = 144\ 031_8 = C819_{16}$
7. $33_{10} = 0010\ 0001_2 = 41_8 = 21_{16}$
8. $19875_{10} = 0100\ 1101\ 1010\ 0011_2 = 46\ 643_8 = 4DA3_{16}$
9. $100125_{10} = 0001\ 1000\ 0111\ 0001\ 1101_2 = 303\ 435_8 = 1871D_{16}$
10. $18_{10} = 0001\ 0010_2 = 22_8 = 12_{16}$
11. $700_{10}$
12. $1285_{10}$
13. $321_{10}$
14. $65625_{10}$
15. $420_{10}$
16. $99_{10}$
17. $501875_{10}$
18. $11_{10}$
19. $444_{10}$
20. $295_{10}$

## 3 Conversiones Rápidas

1. Convertir $11101001101101_2$ a hexadecimal.= $3ABD_{16}$
2. Convertir $ACE_{16}$ a binario. = $1010\ 1100\ 1110_2$
3. Convertir $100111010_2$ a octal. = $274_8$

## ...vertir de decimal a binario / octal / hexadecimal:

**① $237\,625 = 0011\ 1010\ 0000\ 0011\ 1001_2$ | $720071_8$ | $3A039_{16}$**

**binario:**

| 19 18 17 16 | 15 14 13 12 | 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |
|---|---|---|---|---|
| 0 0 1 1 | 1 0 1 0 | 0 0 0 0 | 0 0 1 1 | 1 0 0 1 |

$237\,625 = 2^{17} + 2^{16} + 2^{15} + 2^{13} + 2^5 + 2^4 + 2^3 + 2^0$

$\therefore 237\,625_{10} = 0011\ 1010\ 0000\ 0011\ 1001_2$

**Octal:**

237 625/8 (1)
29 703/8 (7)
3 712/8 (0)
464/8 (0)
58/8 (2)
7 //

$\therefore 237\,625_{10} = 720071_8$

**hexadecimal:**

237 625/16 (9)
14 851/16 (3)
928/16 (0)
58/16 (10=A)
3 //

$\therefore 237\,625_{10} = 3A039_{16}$

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**② $153\,375_{10} = 0010\ 0101\ 0111\ 0001\ 1111_2$ | $453\,437_8$ | $2571F_{16}$**

**binario:**

$153\,375 = 2^{17} + 2^{14} + 2^{12} + 2^{10} + 2^9 + 2^8 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$

| 19 18 17 16 | 15 14 13 12 | 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |
|---|---|---|---|---|
| 0 0 1 0 | 0 1 0 1 | 0 1 1 1 | 0 0 0 1 | 1 1 1 1 |

$\therefore 153\,375_{10} = 0010\ 0101\ 0111\ 0001\ 1111_2$

**Octal:**

153 375/8 (7)
19 171/8 (3)
2396/8 (4)
299/8 (3)
37/8 (5)
4 //

$\therefore 153\,375_{10} = 453\,437_8$

**hexadecimal:**

153 375/16 (15=F)
9585/16 (1)
599/16 (7)
37/16 (5)
2 //

$\therefore 153\,375_{10} = 2571F_{16}$

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**③ $77_{10} = 0100\ 1101_2$ | $115_8$ | $4D_{16}$**

**binario:**

| 7 6 5 4 | 3 2 1 0 |
|---|---|
| 0 1 0 0 | 1 1 0 1 |

$77 = 2^6 + 2^3 + 2^2 + 2^0$

$\therefore 77_{10} = 0100\ 1101_2$

**Octal:**

77/8 (5)
9/8 (1)
1 //

$\therefore 77_{10} = 115_8$

**hexadecimal:**

77/16 (13)
4 //
13=D

$\therefore 77_{10} = 4D_{16}$

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**④ $4205_{10} = 0001\ 0000\ 0110\ 1101_2$ | $10155_8$ | $106D_{16}$**

**binario:**

$4205 = 2^{12} + 2^6 + 2^5 + 2^3 + 2^2 + 2^0$

| 15 14 13 12 | 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |
|---|---|---|---|
| 0 0 0 1 | 0 0 0 0 | 0 1 1 0 | 1 1 0 1 |

$\therefore 4205_{10} = 0001\ 0000\ 0110\ 1101_2$

**Octal:**

4205/8 (5)
525/8 (5)
65/8 (1)
8/8 (0)
1 //

$\therefore 4205_{10} = 10155_8$

**hexadecimal:**

4205/16 (13)
262/16 (6)
16/16 (0)
1 //
13=D

$\therefore 4205_{10} = 106D_{16}$

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**⑤ $255 = 1111\ 1111_2$ | $377_8$ | $FF_{16}$**

**binario:**

$255 = 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$

| 7 6 5 4 | 3 2 1 0 |
|---|---|
| 1 1 1 1 | 1 1 1 1 |

$\therefore 255_{10} = 1111\ 1111_2$

**Octal:**

255/8 (7)
31/8 (7)
3 //

$\therefore 255_{10} = 377_8$

**hexadecimal:**

255/16 (15)
15 //
15=F

$\therefore 255_{10} = FF_{16}$

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**⑥ $51\,225 = 1100\ 1000\ 0001\ 1001_2$ | $144\,031_8$ | $C819_{16}$**

**binario:**

$51\,225 = 2^{15} + 2^{14} + 2^{11} + 2^4 + 2^3 + 2^0$

| 15 14 13 12 | 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |
|---|---|---|---|
| 1 1 0 0 | 1 0 0 0 | 0 0 0 1 | 1 0 0 1 |

$\therefore 51\,225_{10} = 1100\ 1000\ 0001\ 1001_2$

**Octal:**

51 225/8 (1)
6403/8 (3)
800/8 (0)
100/8 (4)
12/8 (4)
1 //

$\therefore 51\,225_{10} = 144031_8$

**hexadecimal:**

51 225/16 (9)
3201/16 (1)
200/16 (8)
12 //
12=C

$\therefore 51\,225_{10} = C819_{16}$

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**⑦ $33_{10} = 0010\ 0001_2$ | $41_8$ | $21_{16}$**

**binario:**

$33 = 2^5 + 2^0$

| 7 6 5 4 | 3 2 1 0 |
|---|---|
| 0 0 1 0 | 0 0 0 1 |

**Octal:**

33/8 (1)
4 //

**hexadecimal:**

33/16
2 //

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**⑧ $19\,875_{10} = 0100\ 1101\ 1010\ 0011_2$ | $46643_8$ | $4DA3_{16}$**

**binario:**

$19875 = 2^{14} + 2^{11} + 2^{10} + 2^8 + 2^7 + 2^5 + 2^1 + 2^0$

| 15 14 13 12 | 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |
|---|---|---|---|
| 0 1 0 0 | 1 1 0 1 | 1 0 1 0 | 0 0 1 1 |

$\therefore 19\,875_{10} = 0100\ 1101\ 1010\ 0011_2$

**Octal:**

19 875/8 (3)
2484/8 (4)
310/8 (6)
38/8 (6)
4 //

$\therefore 19875_{10} = 46643_8$

**hexadecimal:**

19875/16 (3)
1242/16 (10)
77/16 (13)
4 //
10=A
13=D

$\therefore 19875_{10} = 4DA3_{16}$

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| Potencia 2: | $2^{19}$ | $2^{18}$ | $2^{17}$ | $2^{16}$ | $2^{15}$ | $2^{14}$ | $2^{13}$ | $2^{12}$ | $2^{11}$ | $2^{10}$ | $2^9$ | $2^8$ | $2^7$ | $2^6$ | $2^5$ | $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 524 288 | 262 144 | 131 072 | 65 536 | 32 768 | 16 384 | 8192 | 4096 | 2048 | 1024 | 512 | 256 | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |

(9) $100\,125 = 0001\,1000\,0111\,0001\,1101_2 \mid 303\,435_8 \mid 1871D_{16}$

**binario:**

$100\,125 = 2^{16} + 2^{15} + 2^{10} + 2^9 + 2^8 + 2^4 + 2^3 + 2^2 + 2^0$

| 19 18 17 16 | 15 14 13 12 | 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |
|---|---|---|---|---|
| 0001 | 1000 | 0111 | 0001 | 1101 |

$\therefore 100125_{10} = 0001\,1000\,0111\,0001\,1101_2$

**Octal:**

100 125/8 (5)
12 515/8 (3)
1564/8 (4)
195/8 (3)
24/8 (0)
3

$\therefore 100125_{10} = 303\,435_8$

**hexadecimal:**

100125/16 (13)
6257/16 (1)
391/16 (7)
24/16 (8)
1 //

13 = D

$\therefore 100125_{10} = 1871D_{16}$

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(10) $18_{10} = 0001\,0010_2 \mid 22_8 \mid 12_{16}$

**binario:**

$18 = 2^4 + 2^1$

| 7 6 5 4 | 3 2 1 0 |
|---|---|
| 0001 | 0010 |

$\therefore 18_{10} = 0001\,0010_2$

**Octal:**

18/8 (2)
2 //

$\therefore 18_{10} = 22_8$

**hexadecimal:**

18/16 (2)
1 //

$\therefore 18_{10} = 12_{16}$

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## C. Conversiones Rápidas:

(1) $1110\,1001\,1011\,01_2$ a hexadecimal: $3ABD_{16}$

| 0011 | 1010 | 1011 | 1101 |
|---|---|---|---|
| 3 | 10 (A) | 11 (B) | 13 (D) |

$\therefore 1110\,1001\,1011\,01_2 = 3ABD_{16}$

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(2) $ACE_{16}$ a binario: $1010\,1100\,1110_2$

$A = 10_{10} = 1010_2\ (2^3 + 2^1)$
$C = 12_{10} = 1100_2\ (2^3 + 2^2)$
$E = 14_{10} = 1110_2\ (2^3 + 2^2 + 2^1)$

$\therefore ACE_{16} = 1010\,1100\,1110_2$

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(3) $100111010_2$ a Octal: $274_8$

| 100 | 111 | 010 |
|---|---|---|

$010 = 2^1 = 2$
$111 = 2^0 + 2^1 + 2^2 = 7$
$100 = 2^2 = 4$

$\therefore 100111010_2 = 274_8$

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(4) $725_8$ a binario: $0001\,1101\,0101_2$

$7 = 2^2 + 2^1 + 2^0 = 111$
$2 = 2^1 = 010$
$5 = 2^2 + 2^0 = 101$

$\therefore 725_8 = 111\,010\,101_2$

ó ~~0001 1101 0101~~ $0001\,1101\,0101_2$

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(5) $1111\,11111_2$ a hexadecimal: $1FF_{16}$

$0001 = 2^0 = 1$
$1111 = 2^0 + 2^1 + 2^2 + 2^3 = 15 = F$
$1111 = 2^0 + 2^1 + 2^2 + 2^3 = 15 = F$

$\therefore 1111\,11111_2 = 1FF_{16}$

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(6) $2F_{16}$ a binario: $0010\,1111_2$

$F = 15 = 2^0 + 2^1 + 2^2 + 2^3 = 1111_2$
$2 = 2^1 = 0010$

$\therefore 2F_{16} = 0010\,1111_2$

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(7) $1101\,0110_2$ a Octal: $326_8$

| 011 | 010 | 110 |
|---|---|---|

$011 = 2^1 + 2^0 = 3$
$010 = 2^1 = 2$
$110 = 2^2 + 2^1 = 6$

$\therefore 1101\,0110_2 = 326_8$

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(8) $1023_{10}$ a hexadecimal: $3FF_{16}$

1023/16 (15) = F
63/16 (15) = F
3 //

$\therefore 1023_{10} = 3FF_{16}$

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(9) $11101010_2$ a hexadecimal: $166_8$

| 0111 | 0110 |
|---|---|

$0111 = 2^2 + 2^1 + 2^0 = 7$
$0110 = 2^2 + 2^1 = 6$

$\therefore 11101110_2 = 76_{16}$

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(10) $DF_{16}$ a binario = $1101\,1111_2$

$D = 13_{10} = 2^3 + 2^2 + 2^0 = 1101$
$F = 15_{10} = 2^3 + 2^2 + 2^1 + 2^0 = 1111$

$\therefore DF_{16} = 1101\,1111_2$

4. Convertir $725_8$ a binario. = 0001 1101 0101$_2$ = 11101010$_2$
5. Convertir $111111111_2$ a hexadecimal. = 1FF$_{16}$
6. Convertir $2F_{16}$ a binario. = 0010 1111$_2$
7. Convertir $11010110_2$ a octal. = 326$_8$
8. Convertir $1023_{10}$ a hexadecimal. = 3FF$_{16}$
9. Convertir $1110110_2$ a hexadecimal. = 76$_{16}$
10. Convertir $DF_{16}$ a binario. = 1101 1111$_2$
11. Convertir $1001_2$ a hexadecimal.
12. Convertir $700_8$ a binario.
13. Convertir $1F3_{16}$ a binario.
14. Convertir $11110000_2$ a octal.
15. Convertir $F0_{16}$ a binario.
16. Convertir $100100100_2$ a octal.
17. Convertir $377_8$ a binario.
18. Convertir $2E_{16}$ a binario.
19. Convertir $10101010_2$ a hexadecimal.
20. Convertir $FF_{16}$ a binario.